NFH-Kuza-Google — Beckn Schema Definitions (Multi-Domain)

Extension schemas for `local-services` (Agri/Rural) AND `health-services` (TIBU/Health) domains

BECKN DOMAINS	PROTOCOL VERSION	EXTENSION SCHEMAS	SUB-SCHEMAS
`local-services` · `health-services`	Beckn v2.0.0	14 schemas (7 + 7)	4 shared · 2 health-specific

This document defines the Beckn v2.0.0 extension schemas used by the NFH-Kuza-Google platform across **two Beckn domains**:

- **`local-services`** — the existing agri/rural domain covering livelihood services, household services, market produce aggregation, and advisory content.
- **`health-services`** — the new health domain (TIBU integration) covering TB screening, TB treatment, supervision visits, and post-TB care / health advisory.

Both domains run on a **shared flow-engine architecture**. Common patterns — the generic `subCategory: {code, name}` structure, capability matching, and the unified service status lifecycle — appear in both, differing only in the domain-specific values and sub-schemas. Shared sub-schemas (`ServiceArea`, `QuantitySpec`, `Feedback`, and the generic seeker location) are reused across both domains.

> **Sub-categories are dynamic.** Within each category, sub-categories are managed as master data in the Kuza Platform and are not hardcoded in these schemas. The `subCategory` field uses a generic `{code, name}` structure to support any sub-categorization.

> **Choosing a schema set.** See the Domain Selection Guide below to decide which Beckn domain and itemAttributes schema applies to your use case.

Domain Selection Guide

Use the table below to pick the correct Beckn domain and itemAttributes extension for a given use case.

USE CASE	BECKN DOMAIN	ITEMATTRIBUTES SCHEMA
Farmer books a vet for vaccination	`local-services`	`LivelihoodItemAttributes`
Patient gets TB screening at facility	`health-services`	`HealthScreeningItemAttributes`
Household pays school fees	`local-services`	`HouseholdItemAttributes`
Patient on TB treatment (DSTB/DRTB/TPT)	`health-services`	`HealthTreatmentItemAttributes`
Farmer sells eggs / produce to aggregator	`local-services`	`MarketProduceItemAttributes`
SCTLC / CTLC visits facility for supervision	`health-services`	`HealthSupervisionItemAttributes`
Farmer gets agri advisory video	`local-services`	`AdvisoryContentItemAttributes`
Patient gets TB awareness / adherence content	`health-services`	`HealthAdvisoryItemAttributes`

Category Codes — `local-services`

`Livelihood-Services`
LIVELIHOOD FLOW
Any income/livelihood-related service or input (veterinary, crop, trades/skills services)

`Household-Services`
HOUSEHOLD FLOW
Any household need (education, health, insurance, consumer goods)

`Market-Produce`
MARKET FLOW
Aggregation and sale of any goods/produce. Farmer is the seller, provider is the buyer/aggregator.

`Advisory-Content`
ADVISORY FLOW
Any informational/training content delivery. Free, discover-only — no order lifecycle.

CATEGORY CODE	FLOW	HAS ORDER LIFECYCLE	PAYMENT
`Livelihood-Services`	Livelihood	Yes	Paid
`Household-Services`	Household	Yes	Paid
`Market-Produce`	Market	Yes	Paid
`Advisory-Content`	Advisory	No — discover only	Free

Category Codes — `health-services`

`TB-Screening`
SCREENING FLOW
Symptom screening, sputum collection, X-ray, contact tracing (e.g., GeneXpert)

`TB-Treatment`
TREATMENT FLOW
DSTB / DRTB / TPT regimens with adherence tracking and ADR reporting

`TB-Supervision`
SUPERVISION FLOW
SCTLC / CTLC visits, data quality audits, mortality audits, mentorship

`Post-TB-Care`
ADVISORY / FOLLOW-UP FLOW
TB awareness, adherence education, side-effects management content

Beckn v2 Core Entity Mapping

Both domains attach their extension schemas to the same set of Beckn v2 containers. The container is identical; only the extension schema plugged into it changes per domain.

EXTENSION CONTAINER	BECKN V2 CORE ENTITY	`LOCAL-SERVICES` SCHEMAS	`HEALTH-SERVICES` SCHEMAS
`itemAttributes`	Resource / Item	LivelihoodItemAttributes, HouseholdItemAttributes, MarketProduceItemAttributes, AdvisoryContentItemAttributes	HealthScreeningItemAttributes, HealthTreatmentItemAttributes, HealthSupervisionItemAttributes, HealthAdvisoryItemAttributes
`offerAttributes`	Offer	LocalServiceOfferAttributes	HealthServiceOfferAttributes
`fulfillmentAttributes`	Performance / Fulfillment	LocalServiceFulfillmentAttributes	HealthServiceFulfillmentAttributes
`orderAttributes`	Contract / Order	LocalServiceOrderAttributes	HealthServiceOrderAttributes

Schema selection is driven by domain and category. Both domains use the same four Beckn containers — just with domain-specific extensions plugged in. A `local-services` message for `Livelihood-Services` carries `LivelihoodItemAttributes`; a `health-services` message for `TB-Screening` carries `HealthScreeningItemAttributes`. Advisory categories in either domain may use only `itemAttributes` and `fulfillmentAttributes`.

Section A — `local-services` Domain Schemas

The 7 extension schemas below apply to the `local-services` Beckn domain covering Livelihood, Household, Market-Produce, and Advisory flows.

1. LivelihoodItemAttributes

`lia:` **LivelihoodItemAttributes**

Container: **itemAttributes** Applies to: **Livelihood-Services**

Describes a livelihood-related service or input that a farmer is seeking. Covers veterinary services, crop services, input supply, and trades/skills.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping within Livelihood	`{"code": "livestock-services", "name": "Livestock Services"}`
`serviceName`	`string`	REQUIRED	Human-readable name of the service	`"Poultry Vaccination"`
`serviceDescription`	`string`	OPTIONAL	Detailed description of the service	`"Newcastle Disease vaccination for broiler chickens"`
`serviceMode`	`enum`	REQUIRED	How the service is delivered. Values: ON_SITE_VISIT PROVIDER_LOCATION INPUT_DELIVERY	`"ON_SITE_VISIT"`
`quantityRequired`	`QuantitySpec`	OPTIONAL	What the farmer needs serviced/supplied	`{"value": 200, "unit": "birds", "description": "Birds to vaccinate"}`
`estimatedDuration`	`string` ISO 8601	OPTIONAL	Estimated time to deliver the service	`"PT2H"`
`inputDetails`	`object`	OPTIONAL	For input supply: {inputName, unit, quantityPerUnit}	`{"inputName": "Broiler Feed", "unit": "Kg", "quantityPerUnit": 50}`
`sectorContext`	`object`	OPTIONAL	Domain-specific context: {livestockType, breed, cropType, tradeType}	`{"livestockType": "Poultry", "breed": "Broiler"}`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags for discoverability	`["vaccination", "poultry", "broiler"]`

OpenAPI 3.1 Definition

```yaml
# LivelihoodItemAttributes (lia:)
LivelihoodItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - serviceMode
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code:
          type: string
          description: Domain grouping code within Livelihood
          examples: ["livestock-services", "crop-services", "trades-skills"]
        name:
          type: string
          description: Human-readable sub-category name
    serviceName:
      type: string
      description: Human-readable name of the service
    serviceDescription:
      type: string
      description: Detailed description of the service
    serviceMode:
      type: string
      enum: [ON_SITE_VISIT, PROVIDER_LOCATION, INPUT_DELIVERY]
      description: How the service is delivered
    quantityRequired:
      $ref: "#/components/schemas/QuantitySpec"
```

```yaml
    estimatedDuration:
      type: string
      format: duration
      description: ISO 8601 duration
    inputDetails:
      type: object
      properties:
        inputName:
          type: string
        unit:
          type: string
        quantityPerUnit:
          type: number
    sectorContext:
      type: object
      properties:
        livestockType:
          type: string
        breed:
          type: string
        cropType:
          type: string
        tradeType:
          type: string
    availableLanguages:
      type: array
      items:
        type: string
        pattern: "^[a-z]{2}$"
    tags:
      type: array
      items:
        type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-001",
        "descriptor": { "name": "Dr. Wanjiku Veterinary Services" },
        "items": [{
          "id": "item-uuid-001",
          "descriptor": { "name": "Poultry Vaccination" },
          "category_ids": ["Livelihood-Services"],
          "itemAttributes": {
            "subCategory": { "code": "livestock-services", "name": "Livestock Services" },
            "serviceName": "Poultry Vaccination",
            "serviceDescription": "Newcastle Disease vaccination for broiler chickens",
            "serviceMode": "ON_SITE_VISIT",
            "quantityRequired": { "value": 200, "unit": "birds", "description": "Birds to vaccinate" },
            "estimatedDuration": "PT2H",
            "sectorContext": { "livestockType": "Poultry", "breed": "Broiler" },
            "availableLanguages": ["en", "sw"],
            "tags": ["vaccination", "poultry", "newcastle"]
          }
        }]
      }]
    }
  }
}
```

2. HouseholdItemAttributes

hia: HouseholdItemAttributes

Container: **itemAttributes** Applies to: **Household-Services**

Describes a household-level need: education, health, insurance, consumer goods, or any non-livelihood service.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping within Household	`{"code": "education", "name": "Education"}`
`serviceName`	`string`	REQUIRED	Human-readable name	`"School Enrollment Assistance"`
`serviceDescription`	`string`	OPTIONAL	Detailed description	`"Help with primary school enrollment forms and fee payment"`
`serviceMode`	`enum`	REQUIRED	How the service is delivered. Values: SERVICE_DELIVERY PRODUCT_DELIVERY FACILITY_VISIT DIGITAL	`"SERVICE_DELIVERY"`
`quantityRequired`	`QuantitySpec`	OPTIONAL	What the household needs	`{"value": 2, "unit": "children", "description": "Children to enroll"}`
`beneficiaryType`	`enum`	OPTIONAL	Who the service is for. Values: INDIVIDUAL HOUSEHOLD CHILD ELDERLY	`"CHILD"`
`documentsRequired`	`string[]`	OPTIONAL	Documents the household needs to provide	`["Birth Certificate", "National ID"]`
`serviceFrequency`	`enum`	OPTIONAL	How often the service occurs. Values: ONE_TIME RECURRING SEASONAL	`"ONE_TIME"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags for discoverability	`["education", "enrollment", "primary-school"]`

OpenAPI 3.1 Definition

```yaml
# HouseholdItemAttributes (hia:)
HouseholdItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - serviceMode
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code:
          type: string
          examples: ["education", "health", "insurance", "consumer-goods"]
        name:
          type: string
    serviceName:
      type: string
    serviceDescription:
      type: string
    serviceMode:
      type: string
      enum: [SERVICE_DELIVERY, PRODUCT_DELIVERY, FACILITY_VISIT, DIGITAL]
    quantityRequired:
      $ref: "#/components/schemas/QuantitySpec"
```

```yaml
    beneficiaryType:
      type: string
      enum: [INDIVIDUAL, HOUSEHOLD, CHILD, ELDERLY]
    documentsRequired:
      type: array
      items:
        type: string
    serviceFrequency:
      type: string
      enum: [ONE_TIME, RECURRING, SEASONAL]
    availableLanguages:
      type: array
      items:
        type: string
        pattern: "^[a-z]{2}$"
    tags:
      type: array
      items:
        type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-020",
        "descriptor": { "name": "Kiambu Education Services" },
        "items": [{
          "id": "item-uuid-020",
          "descriptor": { "name": "School Enrollment Assistance" },
          "category_ids": ["Household-Services"],
          "itemAttributes": {
            "subCategory": { "code": "education", "name": "Education" },
            "serviceName": "School Enrollment Assistance",
            "serviceDescription": "Help with primary school enrollment forms and fee payment",
            "serviceMode": "SERVICE_DELIVERY",
            "beneficiaryType": "CHILD",
            "documentsRequired": ["Birth Certificate", "National ID"],
            "serviceFrequency": "ONE_TIME",
            "availableLanguages": ["en", "sw"],
            "tags": ["education", "enrollment"]
          }
        }]
      }]
    }
  }
}
```

3. MarketProduceItemAttributes

`mpia:` **MarketProduceItemAttributes**

Container: **itemAttributes** Applies to: **Market-Produce**

Describes the aggregation/buying service. In this flow the farmer is the seller and the provider is the buyer/aggregator.

> **Inverted roles.** Unlike other categories where the provider delivers a service to the farmer, in Market-Produce the "item" is the aggregation/buying service. The provider (aggregator) buys produce from the farmer.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping within Market	`{"code": "livestock-products", "name": "Livestock Products"}`
`serviceName`	`string`	REQUIRED	Human-readable name	`"Egg Collection Service"`
`serviceDescription`	`string`	OPTIONAL	Detailed description	`"Weekly egg collection from farm gate"`
`produceCategory`	`string`	REQUIRED	What the aggregator buys	`"Eggs"`
`acceptedGrades`	`string[]`	OPTIONAL	Quality grades the aggregator accepts	`["Grade A", "Grade B"]`
`unitOfMeasure`	`string`	REQUIRED	Pricing/quantity unit	`"Tray (30 eggs)"`
`minimumQuantity`	`number`	OPTIONAL	Minimum lot size the aggregator will collect	`5`
`quantityAvailable`	`object` {value, unit}	OPTIONAL	What the farmer has available to sell	`{"value": 10, "unit": "Trays"}`
`collectionMode`	`enum`	REQUIRED	How produce is collected. Values: FARM_GATE_PICKUP COLLECTION_POINT FARMER_DELIVERS	`"FARM_GATE_PICKUP"`
`paymentTimeline`	`enum`	OPTIONAL	When farmer gets paid. Values: IMMEDIATE UPON_COLLECTION AFTER_SALE	`"UPON_COLLECTION"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["eggs", "poultry-products", "collection"]`

OpenAPI 3.1 Definition

```yaml
# MarketProduceItemAttributes (mpia:)
MarketProduceItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - produceCategory
    - unitOfMeasure
    - collectionMode
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code:
          type: string
          examples: ["livestock-products", "farm-produce", "handicrafts"]
        name:
          type: string
    serviceName:
      type: string
    serviceDescription:
      type: string
    produceCategory:
      type: string
      description: What the aggregator buys
    acceptedGrades:
      type: array
      items:
```

```yaml
        type: string
    unitOfMeasure:
      type: string
      description: Pricing/quantity unit
    minimumQuantity:
      type: number
      description: Minimum lot size
    quantityAvailable:
      type: object
      properties:
        value:
          type: number
        unit:
          type: string
    collectionMode:
      type: string
      enum: [FARM_GATE_PICKUP, COLLECTION_POINT, FARMER_DELIVERS]
    paymentTimeline:
      type: string
      enum: [IMMEDIATE, UPON_COLLECTION, AFTER_SALE]
    availableLanguages:
      type: array
      items:
        type: string
        pattern: "^[a-z]{2}$"
    tags:
      type: array
      items:
        type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-030",
        "descriptor": { "name": "Kienyeji Egg Traders Ltd" },
        "items": [{
          "id": "item-uuid-030",
          "descriptor": { "name": "Egg Collection Service" },
          "category_ids": ["Market-Produce"],
          "itemAttributes": {
            "subCategory": { "code": "livestock-products", "name": "Livestock Products" },
            "serviceName": "Egg Collection Service",
            "produceCategory": "Eggs",
            "acceptedGrades": ["Grade A", "Grade B"],
            "unitOfMeasure": "Tray (30 eggs)",
            "minimumQuantity": 5,
            "quantityAvailable": { "value": 10, "unit": "Trays" },
            "collectionMode": "FARM_GATE_PICKUP",
            "paymentTimeline": "UPON_COLLECTION",
            "tags": ["eggs", "poultry-products"]
          }
        }]
      }]
    }
  }
}
```

4. AdvisoryContentItemAttributes

acia: AdvisoryContentItemAttributes

Container: **itemAttributes** Applies to: **Advisory-Content**

Describes free advisory/training content. No order lifecycle — content URL is delivered directly in `on_discover`.

> **Advisory is discover-only.** There is no select/init/confirm flow for advisory content. The `contentUrl` is delivered directly in the `on_discover` response, and no `offerAttributes` or `orderAttributes` are used.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Topic grouping within Advisory	`{"code": "agricultural-advisory", "name": "Agricultural Advisory"}`
`serviceName`	`string`	REQUIRED	Content title	`"Broiler Feeding Schedule - Week 1"`
`serviceDescription`	`string`	OPTIONAL	Content description	`"Optimal feeding schedule and quantities for week-1 broiler chicks"`
`contentType`	`enum`	REQUIRED	Media type. Values: TEXT VIDEO PDF AUDIO IMAGE INFOGRAPHIC	`"VIDEO"`
`contentMimeType`	`string`	OPTIONAL	IANA MIME type	`"video/mp4"`
`contentUrl`	`string` (uri)	REQUIRED	Direct URL to the content	`"https://platformcdn.kuza.one/global/content/video/feeding_schedule.mp4"`
`contentDuration`	`string` ISO 8601	OPTIONAL	Duration for video/audio content	`"PT3M45S"`
`targetAudience`	`string`	OPTIONAL	Who the content is most relevant for	`"Poultry farmers in first cycle"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["feeding", "broiler", "week-1"]`

OpenAPI 3.1 Definition

```yaml
# AdvisoryContentItemAttributes (acia:)
AdvisoryContentItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - contentType
    - contentUrl
```

```yaml
    properties:
      subCategory:
        type: object
        required: [code, name]
        properties:
          code:
            type: string
            examples: ["agricultural-advisory", "financial-literacy", "health-nutrition"]
          name:
            type: string
      serviceName:
        type: string
        description: Content title
      serviceDescription:
        type: string
      contentType:
        type: string
        enum: [TEXT, VIDEO, PDF, AUDIO, IMAGE, INFOGRAPHIC]
      contentMimeType:
        type: string
        description: IANA MIME type
      contentUrl:
        type: string
        format: uri
        description: Direct URL to content
      contentDuration:
        type: string
        format: duration
        description: ISO 8601 duration for video/audio
      targetAudience:
        type: string
      availableLanguages:
        type: array
        items:
          type: string
          pattern: "^[a-z]{2}$"
      tags:
        type: array
        items:
          type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-kuza",
        "descriptor": { "name": "Kuza Advisory Platform" },
        "items": [{
          "id": "item-uuid-040",
          "descriptor": { "name": "Broiler Feeding Schedule - Week 1" },
          "category_ids": ["Advisory-Content"],
          "itemAttributes": {
            "subCategory": { "code": "agricultural-advisory", "name": "Agricultural Advisory" },
            "serviceName": "Broiler Feeding Schedule - Week 1",
            "serviceDescription": "Optimal feeding schedule and quantities for week-1 broiler chicks",
            "contentType": "VIDEO",
            "contentMimeType": "video/mp4",
            "contentUrl": "https://platformcdn.kuza.one/global/content/video/feeding_wk1.mp4",
            "contentDuration": "PT3M45S",
            "targetAudience": "Poultry farmers in first cycle",
            "availableLanguages": ["en", "sw"],
            "tags": ["feeding", "broiler", "week-1"]
          }
        }]
      }]
    }
```

```
  }
}
```

5. LocalServiceOfferAttributes

lsoa: LocalServiceOfferAttributes

Container: **offerAttributes** Applies to: **Livelihood-Services, Household-Services, Market-Produce**

Pricing, payment methods, availability, and capacity for all paid flows. Not used for Advisory-Content.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`priceAmount`	`number`	REQUIRED	Base price	`50`
`currency`	`string`	REQUIRED	ISO 4217 currency code	`"KES"`
`priceUnit`	`string`	OPTIONAL	What the price is per	`"per bird"`
`pricingModel`	`enum`	REQUIRED	Pricing approach. Values: FIXED PER_UNIT PER_VISIT NEGOTIABLE	`"PER_UNIT"`
`acceptedPaymentMethods`	`enum[]`	REQUIRED	How the provider accepts payment. Values: CASH MPESA	`["CASH", "MPESA"]`
`capacity`	`object`	OPTIONAL	Provider capacity: {value, unit, period}	`{"value": 500, "unit": "birds", "period": "per day"}`
`serviceAvailability`	`object`	OPTIONAL	When the provider is available: {availableDays, timeSlots}	`{"availableDays": ["Mon","Tue","Wed","Thu","Fri"], "timeSlots": ["08:00-12:00","14:00-17:00"]}`
`cancellationPolicy`	`object`	OPTIONAL	Cancellation rules: {isCancellable, windowHours, reasons}	`{"isCancellable": true, "windowHours": 24, "reasons": ["Weather","Emergency"]}`
`validFrom`	`string` (datetime)	OPTIONAL	Offer start date (ISO 8601)	`"2026-04-01T00:00:00Z"`
`validTo`	`string` (datetime)	OPTIONAL	Offer expiry date (ISO 8601)	`"2026-06-30T23:59:59Z"`

OpenAPI 3.1 Definition

```yaml
# LocalServiceOfferAttributes (lsoa:)
LocalServiceOfferAttributes:
  type: object
  required:
    - priceAmount
    - currency
    - pricingModel
    - acceptedPaymentMethods
  properties:
    priceAmount:
      type: number
      description: Base price
    currency:
      type: string
      pattern: "^[A-Z]{3}$"
      description: ISO 4217 currency code
    priceUnit:
```

```yaml
      type: string
      description: What the price is per (e.g., "per bird", "per visit")
    pricingModel:
      type: string
      enum: [FIXED, PER_UNIT, PER_VISIT, NEGOTIABLE]
    acceptedPaymentMethods:
      type: array
      items:
        type: string
        enum: [CASH, MPESA]
      minItems: 1
    capacity:
      type: object
      properties:
        value:
          type: number
        unit:
          type: string
        period:
          type: string
    serviceAvailability:
      type: object
      properties:
        availableDays:
          type: array
          items:
            type: string
        timeSlots:
          type: array
          items:
            type: string
    cancellationPolicy:
      type: object
      properties:
        isCancellable:
          type: boolean
        windowHours:
          type: integer
        reasons:
          type: array
          items:
            type: string
    validFrom:
      type: string
      format: date-time
    validTo:
      type: string
      format: date-time
```

JSON Example — Beckn Message Fragment

```json
{
  "offerAttributes": {
    "priceAmount": 50,
    "currency": "KES",
    "priceUnit": "per bird",
    "pricingModel": "PER_UNIT",
    "acceptedPaymentMethods": ["CASH", "MPESA"],
    "capacity": { "value": 500, "unit": "birds", "period": "per day" },
    "serviceAvailability": {
      "availableDays": ["Mon", "Tue", "Wed", "Thu", "Fri"],
      "timeSlots": ["08:00-12:00", "14:00-17:00"]
    },
    "cancellationPolicy": {
      "isCancellable": true,
      "windowHours": 24,
      "reasons": ["Weather", "Emergency"]
    },
    "validFrom": "2026-04-01T00:00:00Z",
    "validTo": "2026-06-30T23:59:59Z"
  }
}
```

6. LocalServiceFulfillmentAttributes

lsfa: LocalServiceFulfillmentAttributes

Container: **fulfillmentAttributes** Applies to: **All flows** (including Advisory for location context)

Location, scheduling, delivery mode, and service status tracking. Used across all four category flows.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`serviceArea`	`ServiceArea[]`	REQUIRED (PROVIDER)	Provider's registered coverage areas	`[{"countyCode": "022", "countyName": "Kiambu", "subCounties": [{"subCountyCode": "022-01", "subCountyName": "Thika Town"}]}]`
`farmerLocation`	`FarmerLocation`	REQUIRED (FARMER)	Farmer's location details	`{"county": "Kiambu", "subCounty": "Thika Town", "ward": "Kamenu", "village": "Makongeni"}`
`scheduledDate`	`string` (date)	OPTIONAL	Farmer's preferred service date (ISO 8601)	`"2026-04-15"`
`scheduledTimeSlot`	`string`	OPTIONAL	Preferred time of day	`"Morning"`
`confirmedVisitTime`	`string` (datetime)	OPTIONAL	Provider's confirmed visit date/time	`"2026-04-15T09:00:00+03:00"`
`actualServiceDate`	`string` (date)	OPTIONAL	When service was actually provided	`"2026-04-15"`
`deliveryMode`	`enum`	REQUIRED	How the service is delivered. Values: ON_SITE PROVIDER_LOCATION COLLECTION_POINT DIGITAL	`"ON_SITE"`
`serviceStatus`	`enum`	REQUIRED	Current status in lifecycle. Values: ORDER_PLACED PROVIDER_ACCEPTED PROVIDER_REJECTED IN_PROGRESS COMPLETED FARMER_CONFIRMED CANCELLED CLOSED	`"PROVIDER_ACCEPTED"`
`providerNotes`	`string`	OPTIONAL	Notes from the provider's visit	`"Vaccinated 195 birds. 5 birds showed signs of illness, recommended isolation."`
`followUpRequired`	`boolean`	OPTIONAL	Provider flags if follow-up is needed	`true`

OpenAPI 3.1 Definition

```yaml
# LocalServiceFulfillmentAttributes (lsfa:)
LocalServiceFulfillmentAttributes:
  type: object
  required:
    - deliveryMode
    - serviceStatus
  properties:
    serviceArea:
      type: array
      description: Provider's registered coverage areas (required for provider)
      items:
        $ref: "#/components/schemas/ServiceArea"
    farmerLocation:
```

```yaml
        $ref: "#/components/schemas/FarmerLocation"
      scheduledDate:
        type: string
        format: date
        description: Farmer's preferred service date
      scheduledTimeSlot:
        type: string
        description: Preferred time (e.g., "Morning", "Afternoon")
      confirmedVisitTime:
        type: string
        format: date-time
        description: Provider's confirmed date/time
      actualServiceDate:
        type: string
        format: date
        description: When service was actually provided
      deliveryMode:
        type: string
        enum: [ON_SITE, PROVIDER_LOCATION, COLLECTION_POINT, DIGITAL]
      serviceStatus:
        type: string
        enum:
          - ORDER_PLACED
          - PROVIDER_ACCEPTED
          - PROVIDER_REJECTED
          - IN_PROGRESS
          - COMPLETED
          - FARMER_CONFIRMED
          - CANCELLED
          - CLOSED
      providerNotes:
        type: string
      followUpRequired:
        type: boolean
```

JSON Example — Beckn Message Fragment

```json
{
  "fulfillmentAttributes": {
    "serviceArea": [{
      "countyCode": "022",
      "countyName": "Kiambu",
      "subCounties": [{
        "subCountyCode": "022-01",
        "subCountyName": "Thika Town"
      }]
    }],
    "farmerLocation": {
      "county": "Kiambu",
      "subCounty": "Thika Town",
      "ward": "Kamenu",
      "village": "Makongeni",
      "gpsCoordinates": { "latitude": -1.0396, "longitude": 37.0900 }
    },
    "scheduledDate": "2026-04-15",
    "scheduledTimeSlot": "Morning",
    "confirmedVisitTime": "2026-04-15T09:00:00+03:00",
    "deliveryMode": "ON_SITE",
    "serviceStatus": "PROVIDER_ACCEPTED",
    "followUpRequired": false
  }
}
```

7. LocalServiceOrderAttributes

`lsor:` **LocalServiceOrderAttributes**

Container: **orderAttributes** Applies to: **Livelihood-Services, Household-Services, Market-Produce**

Post-transaction attributes for all paid flows. Both sides (farmer and provider) report payment status independently, enabling reconciliation without a shared payment gateway.

Dual-sided reporting. Payment is reported independently by both parties — the farmer reports via WhatsApp and the provider reports via the web app. There is no integrated payment gateway; reconciliation happens by comparing both reports.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`farmerPaymentMethod`	`enum`	OPTIONAL	How the farmer paid. Values: CASH MPESA	`"MPESA"`
`farmerPaymentStatus`	`enum`	OPTIONAL	Farmer's self-reported payment status. Values: PAID NOT_PAID	`"PAID"`
`providerPaymentReceived`	`boolean`	OPTIONAL	Provider confirms receipt of payment	`true`
`providerAmountReceived`	`number`	OPTIONAL	Amount the provider received	`10000`
`paymentCurrency`	`string`	OPTIONAL	ISO 4217 currency code	`"KES"`
`serviceConfirmedByFarmer`	`boolean`	OPTIONAL	Farmer confirms the service was performed	`true`
`providerFulfillmentStatus`	`enum`	OPTIONAL	Provider's self-reported fulfillment status. Values: COMPLETED NOT_COMPLETED	`"COMPLETED"`
`providerFeedback`	`Feedback`	OPTIONAL	Provider's feedback: {rating, selectedOption, additionalText}	`{"rating": 4, "selectedOption": "Good experience", "additionalText": "Farmer was prepared"}`
`farmerFeedback`	`Feedback`	OPTIONAL	Farmer's feedback: {rating, selectedOption, additionalText}	`{"rating": 5, "selectedOption": "Excellent service"}`
`followUpRequired`	`boolean`	OPTIONAL	Provider flags if follow-up is needed	`true`
`followUpNotes`	`string`	OPTIONAL	What follow-up is needed	`"Booster vaccination needed in 2 weeks"`

OpenAPI 3.1 Definition

```yaml
# LocalServiceOrderAttributes (lsor:)
LocalServiceOrderAttributes:
  type: object
  properties:
    farmerPaymentMethod:
      type: string
      enum: [CASH, MPESA]
      description: Reported by farmer via WhatsApp
    farmerPaymentStatus:
      type: string
      enum: [PAID, NOT_PAID]
    providerPaymentReceived:
      type: boolean
      description: Provider confirms payment receipt via App
    providerAmountReceived:
      type: number
    paymentCurrency:
      type: string
      pattern: "^[A-Z]{3}$"
    serviceConfirmedByFarmer:
      type: boolean
```

```
      description: Farmer confirms service was done
    providerFulfillmentStatus:
      type: string
      enum: [COMPLETED, NOT_COMPLETED]
    providerFeedback:
      $ref: "#/components/schemas/Feedback"
    farmerFeedback:
      $ref: "#/components/schemas/Feedback"
    followUpRequired:
      type: boolean
    followUpNotes:
      type: string
```

JSON Example — Beckn Message Fragment

```
{
  "orderAttributes": {
    "farmerPaymentMethod": "MPESA",
    "farmerPaymentStatus": "PAID",
    "providerPaymentReceived": true,
    "providerAmountReceived": 10000,
    "paymentCurrency": "KES",
    "serviceConfirmedByFarmer": true,
    "providerFulfillmentStatus": "COMPLETED",
    "providerFeedback": {
      "rating": 4,
      "selectedOption": "Good experience",
      "additionalText": "Farmer was well-prepared, birds accessible"
    },
    "farmerFeedback": {
      "rating": 5,
      "selectedOption": "Excellent service"
    },
    "followUpRequired": true,
    "followUpNotes": "Booster vaccination needed in 2 weeks"
  }
}
```

Section B — `health-services` Domain Schemas

The 7 extension schemas below apply to the `health-services` Beckn domain (TIBU integration) covering TB screening, treatment, supervision, and post-TB care / health advisory flows. These share the same Beckn container model as `local-services` but introduce health-specific fields (disease targets, regimens, clinical credentials, outcomes).

1. HealthScreeningItemAttributes

`hsia:` HealthScreeningItemAttributes

Container: **itemAttributes** Domain: **health-services** Applies to: **TB-Screening** and other disease-screening categories

Describes a disease-screening service such as TB symptom screening, sputum collection, chest X-ray, or contact tracing. Used when a patient or CHP requests a screening via Beckn discovery.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code,	REQUIRED	Specific screening sub-category	`{"code": "tb-screening", "name": "TB Screening"}`

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
	name}			
screeningType	enum	REQUIRED	Values: SYMPTOM_SCREENING SPUTUM_COLLECTION XRAY_SCREENING CONTACT_TRACING OTHER	"SPUTUM_COLLECTION"
diseaseTarget	string	REQUIRED	Disease being screened for	"Tuberculosis"
diagnosticMethod	string	OPTIONAL	Lab/method used	"GeneXpert"
requiresLabCapability	string[]	OPTIONAL	Lab capabilities required at facility	["GeneXpert"]
patientEligibility	object	OPTIONAL	{minAge, maxAge, riskFactors[]}	{"minAge": 5, "riskFactors": ["HIV+", "Contact"]}
expectedTurnaroundHours	integer	OPTIONAL	Hours to result	48
isFreeAtPointOfCare	boolean	OPTIONAL	Default true for government programs	true
availableLanguages	string[]	OPTIONAL	ISO 639-1 codes	["en", "sw"]
tags	string[]	OPTIONAL	Free-form tags	["tb", "genexpert", "mdr-tb"]

OpenAPI 3.1 Definition

```yaml
# HealthScreeningItemAttributes (hsia:)
HealthScreeningItemAttributes:
  type: object
  required: [subCategory, screeningType, diseaseTarget]
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code: { type: string }
        name: { type: string }
    screeningType:
      type: string
      enum: [SYMPTOM_SCREENING, SPUTUM_COLLECTION, XRAY_SCREENING, CONTACT_TRACING, OTHER]
    diseaseTarget: { type: string }
    diagnosticMethod: { type: string }
    requiresLabCapability:
      type: array
      items: { type: string }
    patientEligibility:
      type: object
      properties:
        minAge: { type: integer }
        maxAge: { type: integer }
        riskFactors:
          type: array
          items: { type: string }
    expectedTurnaroundHours: { type: integer }
    isFreeAtPointOfCare: { type: boolean }
    availableLanguages:
      type: array
      items: { type: string, pattern: "^[a-z]{2}$" }
    tags:
      type: array
      items: { type: string }
```

JSON Example — Beckn Message Fragment

```json
{
  "context": { "domain": "health-services", "action": "on_discover", "version": "2.0.0" },
  "message": {
    "catalog": {
      "providers": [{
        "id": "facility-nrb-001",
```

```
        "descriptor": { "name": "Mbagathi TB Clinic" },
        "items": [{
          "id": "svc-screen-001",
          "descriptor": { "name": "GeneXpert Sputum Test" },
          "category_ids": ["TB-Screening"],
          "itemAttributes": {
            "subCategory": { "code": "tb-screening", "name": "TB Screening" },
            "screeningType": "SPUTUM_COLLECTION",
            "diseaseTarget": "Tuberculosis",
            "diagnosticMethod": "GeneXpert",
            "requiresLabCapability": ["GeneXpert"],
            "expectedTurnaroundHours": 48,
            "isFreeAtPointOfCare": true,
            "availableLanguages": ["en", "sw"],
            "tags": ["tb", "genexpert"]
          }
        }]
      }]
    }
  }
}
```

2. HealthTreatmentItemAttributes

htia: HealthTreatmentItemAttributes

Container: **itemAttributes** Domain: **health-services** Applies to: **TB-Treatment**

Describes a treatment regimen or clinical-care service such as DSTB, DRTB, or TB Preventive Therapy (TPT).

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object`	REQUIRED	Treatment sub-category	`{"code": "dstb-treatment", "name": "DSTB Treatment"}`
`treatmentType`	`enum`	REQUIRED	Values: DSTB DRTB TPT OTHER	`"DSTB"`
`regimenName`	`string`	OPTIONAL	Standard regimen code	`"2HRZE/4HR"`
`treatmentDurationDays`	`integer`	OPTIONAL	Standard duration	`180`
`frequency`	`string`	OPTIONAL	Visit / review cadence	`"Monthly"`
`requiresMonitoring`	`string[]`	OPTIONAL	Required tests/monitoring	`["Sputum culture", "Liver function"]`
`adherenceTracking`	`boolean`	OPTIONAL	DOT or digital adherence required	`true`
`adverseEventReporting`	`boolean`	OPTIONAL	Whether ADR reporting is mandatory	`true`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["dstb", "first-line"]`

OpenAPI 3.1 Definition

```
# HealthTreatmentItemAttributes (htia:)
HealthTreatmentItemAttributes:
  type: object
  required: [subCategory, treatmentType]
  properties:
```

```
    subCategory:
      type: object
      required: [code, name]
      properties:
        code: { type: string }
        name: { type: string }
    treatmentType:
      type: string
      enum: [DSTB, DRTB, TPT, OTHER]
    regimenName: { type: string }
    treatmentDurationDays: { type: integer }
    frequency: { type: string }
    requiresMonitoring:
      type: array
      items: { type: string }
    adherenceTracking: { type: boolean }
    adverseEventReporting: { type: boolean }
    availableLanguages:
      type: array
      items: { type: string, pattern: "^[a-z]{2}$" }
    tags:
      type: array
      items: { type: string }
```

JSON Example

```
{
  "itemAttributes": {
    "subCategory": { "code": "dstb-treatment", "name": "DSTB Treatment" },
    "treatmentType": "DSTB",
    "regimenName": "2HRZE/4HR",
    "treatmentDurationDays": 180,
    "frequency": "Monthly",
    "requiresMonitoring": ["Sputum smear", "Liver function"],
    "adherenceTracking": true,
    "adverseEventReporting": true
  }
}
```

3. HealthSupervisionItemAttributes

hspia: HealthSupervisionItemAttributes

Container: **itemAttributes** Domain: **health-services** Applies to: **TB-Supervision**

Describes supervision, audit, and mentorship visits such as SCTLC / CTLC quarterly visits, data-quality audits, mortality audits, or clinical mentorship.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object`	REQUIRED	Supervision sub-category	`{"code": "facility-supervision", "name": "Facility Supervision"}`
`supervisionType`	`enum`	REQUIRED	Values: SCTLC_VISIT CTLC_VISIT DATA_QUALITY_AUDIT MORTALITY_AUDIT MENTORSHIP	`"SCTLC_VISIT"`
`targetEntity`	`enum`	REQUIRED	Values: FACILITY CHP CLINICIAN LAB	`"FACILITY"`
`frequency`	`string`	OPTIONAL	Cadence	`"Quarterly"`

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`checklistTemplate`	`string`	OPTIONAL	URI to checklist form/template	`"https://tibu.example/checklists/sctlc-q.json"`
`includesPaymentClaim`	`boolean`	OPTIONAL	Whether visit triggers a payment claim	`true`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1	`["en"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["sctlc", "quarterly"]`

OpenAPI 3.1 Definition

```yaml
# HealthSupervisionItemAttributes (hspia:)
HealthSupervisionItemAttributes:
  type: object
  required: [subCategory, supervisionType, targetEntity]
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code: { type: string }
        name: { type: string }
    supervisionType:
      type: string
      enum: [SCTLC_VISIT, CTLC_VISIT, DATA_QUALITY_AUDIT, MORTALITY_AUDIT, MENTORSHIP]
    targetEntity:
      type: string
      enum: [FACILITY, CHP, CLINICIAN, LAB]
    frequency: { type: string }
    checklistTemplate: { type: string, format: uri }
    includesPaymentClaim: { type: boolean }
    availableLanguages:
      type: array
      items: { type: string, pattern: "^[a-z]{2}$" }
    tags:
      type: array
      items: { type: string }
```

JSON Example

```json
{
  "itemAttributes": {
    "subCategory": { "code": "facility-supervision", "name": "Facility Supervision" },
    "supervisionType": "SCTLC_VISIT",
    "targetEntity": "FACILITY",
    "frequency": "Quarterly",
    "includesPaymentClaim": true
  }
}
```

4. HealthAdvisoryItemAttributes

hadia: **HealthAdvisoryItemAttributes**

Container: **itemAttributes** Domain: **health-services** Applies to: **Post-TB-Care** , awareness content

Describes health advisory, education, or awareness content (TB awareness, adherence importance, side-effect management). Analogous to `AdvisoryContentItemAttributes` for agri, but with health-specific targeting.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object`	REQUIRED	Advisory sub-category	`{"code": "tb-awareness", "name": "TB Awareness"}`
`contentType`	`enum`	REQUIRED	Values: TEXT VIDEO PDF AUDIO INFOGRAPHIC INTERACTIVE	`"VIDEO"`
`contentMimeType`	`string`	OPTIONAL	MIME type	`"video/mp4"`
`contentUrl`	`string` (uri)	REQUIRED	URL to content asset	`"https://cdn.kuza.one/tb/adherence.mp4"`
`contentDuration`	`string` (ISO 8601)	OPTIONAL	Playback duration	`"PT3M20S"`
`targetAudience`	`string`	OPTIONAL	Who the content is for	`"Recently diagnosed TB patients"`
`healthTopic`	`string`	OPTIONAL	Clinical topic	`"Adherence importance"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["adherence", "tb"]`

OpenAPI 3.1 Definition

```yaml
# HealthAdvisoryItemAttributes (hadia:)
HealthAdvisoryItemAttributes:
  type: object
  required: [subCategory, contentType, contentUrl]
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code: { type: string }
        name: { type: string }
    contentType:
      type: string
      enum: [TEXT, VIDEO, PDF, AUDIO, INFOGRAPHIC, INTERACTIVE]
    contentMimeType: { type: string }
    contentUrl: { type: string, format: uri }
    contentDuration: { type: string, format: duration }
    targetAudience: { type: string }
    healthTopic: { type: string }
    availableLanguages:
      type: array
      items: { type: string, pattern: "^[a-z]{2}$" }
    tags:
      type: array
      items: { type: string }
```

JSON Example

```json
{
  "itemAttributes": {
    "subCategory": { "code": "tb-awareness", "name": "TB Awareness" },
    "contentType": "VIDEO",
    "contentUrl": "https://cdn.kuza.one/tb/adherence-sw.mp4",
    "contentDuration": "PT3M20S",
    "targetAudience": "Recently diagnosed TB patients",
    "healthTopic": "Adherence importance",
    "availableLanguages": ["sw"]
  }
}
```

5. HealthServiceOfferAttributes

hsoa: HealthServiceOfferAttributes

Container: **offerAttributes** Domain: **health-services**

Pricing and coverage for health services. Most TIBU services are free at point-of-care under the national TB program; paid / insurance-covered offers are also supported.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`priceAmount`	`number`	OPTIONAL	Default 0 for free services	`0`
`currency`	`string`	OPTIONAL	ISO 4217	`"KES"`
`isFreeAtPointOfCare`	`boolean`	OPTIONAL	Default true	`true`
`coveredBy`	`enum`	OPTIONAL	Values: GOVERNMENT_PROGRAM INSURANCE OUT_OF_POCKET NGO_FUNDED GLOBAL_FUND	`"GOVERNMENT_PROGRAM"`
`coverageNote`	`string`	OPTIONAL	Human-readable coverage note	`"Covered under NTLP"`
`acceptedPaymentMethods`	`enum[]`	OPTIONAL	Values: CASH MPESA INSURANCE_CLAIM NONE	`["NONE"]`
`capacity`	`object`	OPTIONAL	`{value, unit, period}`	`{"value": 20, "unit": "tests", "period": "per-day"}`
`serviceAvailability`	`object`	OPTIONAL	`{availableDays[], timeSlots[]}`	—
`validFrom`	`string` (date-time)	OPTIONAL	—	—
`validTo`	`string` (date-time)	OPTIONAL	—	—

OpenAPI 3.1 Definition

```yaml
# HealthServiceOfferAttributes (hsoa:)
HealthServiceOfferAttributes:
  type: object
  properties:
    priceAmount: { type: number }
    currency: { type: string, pattern: "^[A-Z]{3}$" }
    isFreeAtPointOfCare: { type: boolean }
    coveredBy:
      type: string
      enum: [GOVERNMENT_PROGRAM, INSURANCE, OUT_OF_POCKET, NGO_FUNDED, GLOBAL_FUND]
    coverageNote: { type: string }
    acceptedPaymentMethods:
      type: array
      items: { type: string, enum: [CASH, MPESA, INSURANCE_CLAIM, NONE] }
    capacity:
      type: object
      properties:
        value: { type: number }
        unit: { type: string }
        period: { type: string }
    serviceAvailability:
      type: object
      properties:
        availableDays:
          type: array
```

```
          items: { type: string }
        timeSlots:
          type: array
          items: { type: string }
    validFrom: { type: string, format: date-time }
    validTo: { type: string, format: date-time }
```

JSON Example

```
{
  "offerAttributes": {
    "priceAmount": 0,
    "currency": "KES",
    "isFreeAtPointOfCare": true,
    "coveredBy": "GOVERNMENT_PROGRAM",
    "coverageNote": "Covered under NTLP",
    "acceptedPaymentMethods": ["NONE"],
    "capacity": { "value": 20, "unit": "tests", "period": "per-day" }
  }
}
```

6. HealthServiceFulfillmentAttributes

hsfa: HealthServiceFulfillmentAttributes

Container: **fulfillmentAttributes** Domain: **health-services**

Location, scheduling, facility capabilities, and status tracking for health service delivery. Uses an extended status enum that adds an OUTCOME stage for clinical outcomes.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`facilityServiceArea`	`ServiceArea[]`	PROVIDER	Geographic coverage of the facility	—
`patientLocation`	`SeekerLocation`	SEEKER	Patient's geographic location (same structure as generic SeekerLocation)	—
`facilityType`	`enum`	OPTIONAL	Values: DISPENSARY HEALTH_CENTRE HOSPITAL LAB COMMUNITY MOBILE	`"HEALTH_CENTRE"`
`requiredCapabilities`	`string[]`	OPTIONAL	Required facility capabilities	`["GeneXpert", "X-ray"]`
`providerCredentialRequired`	`string`	OPTIONAL	Required clinical credential	`"Clinical Officer"`
`scheduledDate`	`string` (date)	OPTIONAL	—	—
`scheduledTimeSlot`	`string`	OPTIONAL	—	—
`confirmedVisitTime`	`string` (date-time)	OPTIONAL	—	—
`actualServiceDate`	`string` (date)	OPTIONAL	—	—
`deliveryMode`	`enum`	REQUIRED	Values: FACILITY_VISIT HOME_VISIT TELE_HEALTH MOBILE_CLINIC	`"FACILITY_VISIT"`
`serviceStatus`	`enum`	REQUIRED	Extended lifecycle: NEED_REGISTERED SEARCHED SELECTED CONFIRMED DELIVERED RECORDED	`"OUTCOME_RECORDED"`

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
			OUTCOME_PENDING OUTCOME_RECORDED CANCELLED LOST_TO_FOLLOWUP CLOSED	
`providerNotes`	`string`	OPTIONAL	—	—
`followUpRequired`	`boolean`	OPTIONAL	—	—
`referralRequired`	`boolean`	OPTIONAL	Onward referral needed	`true`
`referralTarget`	`ReferralTarget`	OPTIONAL	Referral destination details	—

OpenAPI 3.1 Definition

```yaml
# HealthServiceFulfillmentAttributes (hsfa:)
HealthServiceFulfillmentAttributes:
  type: object
  required: [deliveryMode, serviceStatus]
  properties:
    facilityServiceArea:
      type: array
      items: { $ref: "#/components/schemas/ServiceArea" }
    patientLocation: { $ref: "#/components/schemas/SeekerLocation" }
    facilityType:
      type: string
      enum: [DISPENSARY, HEALTH_CENTRE, HOSPITAL, LAB, COMMUNITY, MOBILE]
    requiredCapabilities:
      type: array
      items: { type: string }
    providerCredentialRequired: { type: string }
    scheduledDate: { type: string, format: date }
    scheduledTimeSlot: { type: string }
    confirmedVisitTime: { type: string, format: date-time }
    actualServiceDate: { type: string, format: date }
    deliveryMode:
      type: string
      enum: [FACILITY_VISIT, HOME_VISIT, TELE_HEALTH, MOBILE_CLINIC]
    serviceStatus:
      type: string
      enum:
        - NEED_REGISTERED
        - SEARCHED
        - SELECTED
        - CONFIRMED
        - DELIVERED
        - RECORDED
        - OUTCOME_PENDING
        - OUTCOME_RECORDED
        - CANCELLED
        - LOST_TO_FOLLOWUP
        - CLOSED
    providerNotes: { type: string }
    followUpRequired: { type: boolean }
    referralRequired: { type: boolean }
    referralTarget: { $ref: "#/components/schemas/ReferralTarget" }
```

JSON Example

```json
{
  "fulfillmentAttributes": {
    "facilityType": "HEALTH_CENTRE",
    "requiredCapabilities": ["GeneXpert"],
    "providerCredentialRequired": "Clinical Officer",
    "scheduledDate": "2026-04-15",
    "deliveryMode": "FACILITY_VISIT",
    "serviceStatus": "CONFIRMED",
    "referralRequired": false
  }
}
```

7. HealthServiceOrderAttributes

hsor: HealthServiceOrderAttributes

Container: **orderAttributes** Domain: **health-services**

Outcome capture, consent, adverse events, and follow-up for a health service. Links to TIBU patient register and clinical records (e.g., FHIR resources).

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`patientUuid`	`string`	OPTIONAL	TIBU patient identifier	`"pt-0001"`
`providerCredentialId`	`string`	OPTIONAL	Provider's professional licence/ID	`"KMPDB-12345"`
`clinicalRecordRef`	`string`	OPTIONAL	Reference to clinical record / FHIR resource	`"Encounter/abc-123"`
`serviceConfirmedByPatient`	`boolean`	OPTIONAL	Patient-side confirmation	`true`
`providerFulfillmentStatus`	`enum`	OPTIONAL	Values: COMPLETED PARTIAL NOT_COMPLETED	`"COMPLETED"`
`outcomeRecorded`	`boolean`	OPTIONAL	Whether final outcome recorded	`true`
`outcomeType`	`enum`	OPTIONAL	TB values: TB_POSITIVE TB_NEGATIVE CURED TREATMENT_COMPLETED DIED LOST_TO_FOLLOWUP TRANSFERRED_OUT TREATMENT_FAILED NOT_EVALUATED	`"CURED"`
`outcomeDate`	`string` (date)	OPTIONAL	When outcome recorded	`"2026-10-15"`
`outcomeNote`	`string`	OPTIONAL	Clinical note on outcome	`"Completed 6-month regimen"`
`adverseEventReported`	`boolean`	OPTIONAL	—	`false`
`adverseEventDetails`	`AdverseEventDetails`	OPTIONAL	See health sub-schemas	—
`followUpRequired`	`boolean`	OPTIONAL	—	`true`
`followUpScheduledDate`	`string` (date)	OPTIONAL	—	`"2026-11-15"`
`consentGiven`	`boolean`	OPTIONAL	DPDP-equivalent consent for data sharing	`true`
`consentTimestamp`	`string` (date-time)	OPTIONAL	When consent was given	`"2026-04-15T10:00:00Z"`

OpenAPI 3.1 Definition

```yaml
# HealthServiceOrderAttributes (hsor:)
HealthServiceOrderAttributes:
  type: object
  properties:
    patientUuid: { type: string }
    providerCredentialId: { type: string }
    clinicalRecordRef: { type: string }
    serviceConfirmedByPatient: { type: boolean }
    providerFulfillmentStatus:
      type: string
      enum: [COMPLETED, PARTIAL, NOT_COMPLETED]
    outcomeRecorded: { type: boolean }
```

```
    outcomeType:
      type: string
      enum: [TB_POSITIVE, TB_NEGATIVE, CURED, TREATMENT_COMPLETED, DIED, LOST_TO_FOLLOWUP, TRANSFERRED_OUT, TREATMENT_FAILED,
NOT_EVALUATED]
    outcomeDate: { type: string, format: date }
    outcomeNote: { type: string }
    adverseEventReported: { type: boolean }
    adverseEventDetails: { $ref: "#/components/schemas/AdverseEventDetails" }
    followUpRequired: { type: boolean }
    followUpScheduledDate: { type: string, format: date }
    consentGiven: { type: boolean }
    consentTimestamp: { type: string, format: date-time }
```

JSON Example

```
{
  "orderAttributes": {
    "patientUuid": "pt-0001",
    "providerCredentialId": "KMPDB-12345",
    "clinicalRecordRef": "Encounter/abc-123",
    "serviceConfirmedByPatient": true,
    "providerFulfillmentStatus": "COMPLETED",
    "outcomeRecorded": true,
    "outcomeType": "CURED",
    "outcomeDate": "2026-10-15",
    "consentGiven": true,
    "consentTimestamp": "2026-04-15T10:00:00Z"
  }
}
```

Shared Sub-schemas

These shared types are referenced across both `local-services` and `health-services` extension schemas.

ServiceArea

Represents a geographic coverage area used by providers to define where they operate.

FIELD	TYPE	REQUIRED	DESCRIPTION
`countyCode`	`string`	REQUIRED	County identifier code
`countyName`	`string`	REQUIRED	County name
`subCounties`	`object[]`	OPTIONAL	Array of {subCountyCode, subCountyName}

```
# ServiceArea sub-schema
ServiceArea:
  type: object
  required: [countyCode, countyName]
  properties:
    countyCode:
      type: string
    countyName:
      type: string
    subCounties:
      type: array
      items:
        type: object
        required: [subCountyCode, subCountyName]
        properties:
          subCountyCode:
            type: string
```

```
          subCountyName:
            type: string
```

SeekerLocation (aliases: `FarmerLocation`, `PatientLocation`)

Represents the seeker's geographic location for service matching and fulfillment. The same structure is referenced as `farmerLocation` in `LocalServiceFulfillmentAttributes` and as `patientLocation` in `HealthServiceFulfillmentAttributes`. `SeekerLocation` is the canonical cross-domain name; the two aliases are kept for backward compatibility.

FIELD	TYPE	REQUIRED	DESCRIPTION
`county`	`string`	REQUIRED	County name
`subCounty`	`string`	REQUIRED	Sub-county name
`ward`	`string`	OPTIONAL	Ward name
`village`	`string`	OPTIONAL	Village name
`gpsCoordinates`	`object`	OPTIONAL	{latitude: number, longitude: number}

```
# SeekerLocation sub-schema (FarmerLocation / PatientLocation alias the same shape)
SeekerLocation:
  type: object
  required: [county, subCounty]
  properties:
    county:
      type: string
    subCounty:
      type: string
    ward:
      type: string
    village:
      type: string
    gpsCoordinates:
      type: object
      properties:
        latitude:
          type: number
        longitude:
          type: number
```

QuantitySpec

A generic quantity descriptor used across item attributes to specify amounts.

FIELD	TYPE	REQUIRED	DESCRIPTION
`value`	`number`	OPTIONAL	Numeric quantity
`unit`	`string`	OPTIONAL	Unit of measure
`description`	`string`	OPTIONAL	Human-readable description

```
# QuantitySpec sub-schema
QuantitySpec:
  type: object
  properties:
    value:
      type: number
    unit:
      type: string
    description:
      type: string
```

Feedback

Feedback from either party (farmer or provider) after service completion.

FIELD	TYPE	REQUIRED	DESCRIPTION
`rating`	`integer`	OPTIONAL	Rating from 1 to 5
`selectedOption`	`string`	OPTIONAL	Pre-defined feedback option
`additionalText`	`string`	OPTIONAL	Free-text additional feedback

```yaml
# Feedback sub-schema
Feedback:
  type: object
  properties:
    rating:
      type: integer
      minimum: 1
      maximum: 5
    selectedOption:
      type: string
    additionalText:
      type: string
```

Health Sub-schemas

These sub-schemas are used only within `health-services` extension schemas.

AdverseEventDetails

Structured description of an adverse drug reaction or clinical event during treatment.

FIELD	TYPE	REQUIRED	DESCRIPTION
`eventType`	`string`	REQUIRED	Name/classification of the event (e.g., "Hepatotoxicity")
`severity`	`enum`	REQUIRED	Values: MILD MODERATE SEVERE LIFE_THREATENING
`onsetDate`	`string` (date)	OPTIONAL	When the event began
`action`	`string`	OPTIONAL	Action taken (e.g., "Drug discontinued")
`outcome`	`string`	OPTIONAL	Resolution status

```yaml
# AdverseEventDetails sub-schema
AdverseEventDetails:
  type: object
  required: [eventType, severity]
  properties:
    eventType: { type: string }
    severity:
      type: string
      enum: [MILD, MODERATE, SEVERE, LIFE_THREATENING]
    onsetDate: { type: string, format: date }
    action: { type: string }
    outcome: { type: string }
```

ReferralTarget

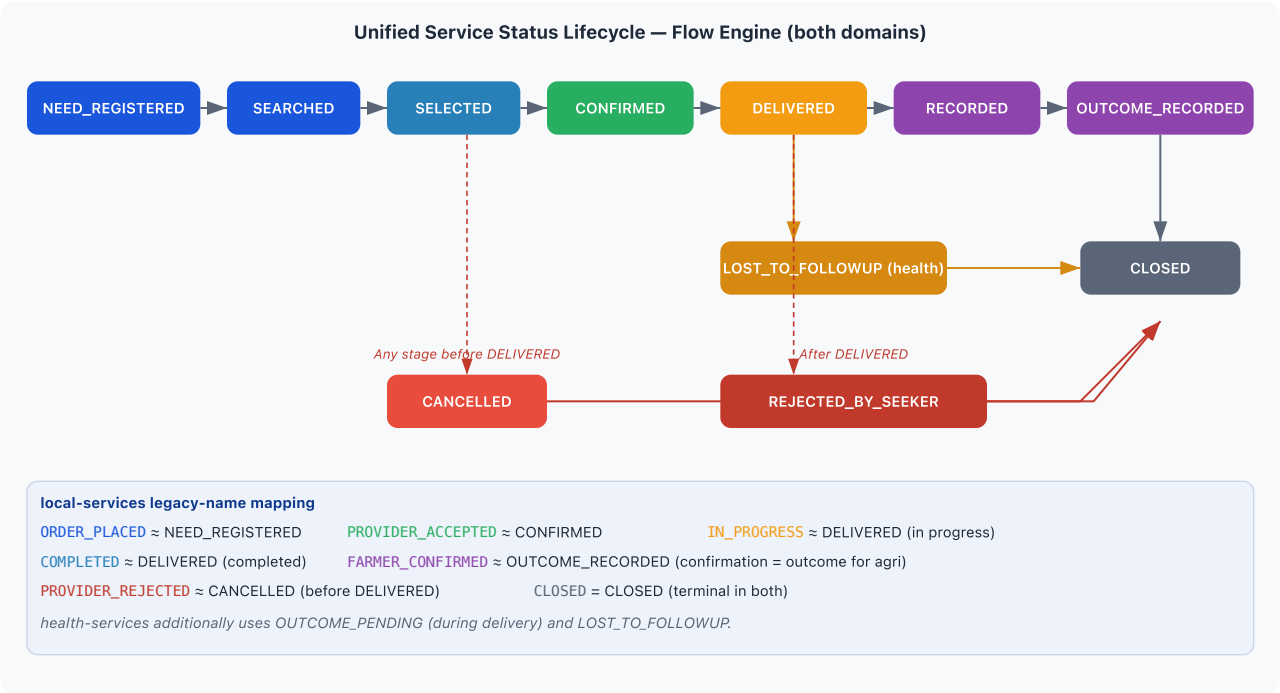
Describes an onward referral destination from a health encounter.

FIELD	TYPE	REQUIRED	DESCRIPTION
`facilityId`	`string`	REQUIRED	Destination facility identifier
`facilityName`	`string`	REQUIRED	Destination facility name
`reason`	`string`	OPTIONAL	Reason for referral
`urgency`	`enum`	OPTIONAL	Values: ROUTINE URGENT EMERGENCY

```yaml
# ReferralTarget sub-schema
ReferralTarget:
  type: object
  required: [facilityId, facilityName]
  properties:
    facilityId: { type: string }
    facilityName: { type: string }
    reason: { type: string }
    urgency:
      type: string
      enum: [ROUTINE, URGENT, EMERGENCY]
```

Unified Service Status Lifecycle

The `serviceStatus` field in the fulfillment extension of either domain tracks the order through a **unified 7-stage flow-engine lifecycle**. `health-services` uses the full enum including the `OUTCOME_*` and `LOST_TO_FOLLOWUP` states. `local-services` uses a compatible subset (legacy names are retained for the existing agri flows and map onto the new states as shown below). Advisory content in either domain has no lifecycle — it is delivered immediately in `on_discover`.



Unified state transition rules:

FROM	TO	TRIGGERED BY	APPLIES TO
`NEED_REGISTERED`	`SEARCHED`	BAP issues discover; catalog returned	both
`SEARCHED`	`SELECTED`	Seeker selects a provider/item	both
`SELECTED`	`CONFIRMED`	Provider accepts the order	both

FROM	TO	TRIGGERED BY	APPLIES TO
`CONFIRMED`	`DELIVERED`	Provider delivers the service	both
`DELIVERED`	`RECORDED`	Provider records completion (notes, data)	both
`RECORDED`	`OUTCOME_RECORDED`	Seeker/patient confirms (agri) OR clinical outcome captured (health)	both
`DELIVERED` / `RECORDED`	`LOST_TO_FOLLOWUP`	Patient cannot be contacted for outcome	`health-services`
`OUTCOME_RECORDED` / `LOST_TO_FOLLOWUP`	`CLOSED`	Transaction finalized	both
Any stage before `DELIVERED`	`CANCELLED`	Either party cancels	both
After `DELIVERED`	`REJECTED_BY_SEEKER`	Seeker disputes / rejects delivered service	both
`CANCELLED` / `REJECTED_BY_SEEKER`	`CLOSED`	Automatic terminal closure	both

Advisory content has no lifecycle in either domain. `AdvisoryContentItemAttributes` (agri) and `HealthAdvisoryItemAttributes` (health) are delivered immediately via `contentUrl` in the `on_discover` response.

Legacy state names retained. Existing `local-services` integrations may continue emitting the legacy enum (`ORDER_PLACED`, `PROVIDER_ACCEPTED`, `IN_PROGRESS`, `COMPLETED`, `FARMER_CONFIRMED`, `PROVIDER_REJECTED`) during the transition period. The flow engine maps these onto the unified states per the diagram above. New integrations should emit the unified names.

Full OpenAPI 3.1 Bundle — `local-services`

The complete `local-services` schema definitions bundled as a single OpenAPI 3.1 components block, suitable for inclusion in a Beckn domain specification file.

```yaml
# ============================================================
# NFH-Kuza-Google Extension Schemas
# Domain: local-services | Protocol: Beckn v2.0.0
# ============================================================

openapi: "3.1.0"
info:
  title: NFH-Kuza-Google Extension Schemas
  version: "1.0.0"
  description: >
    Domain extension schemas for the local-services domain
    in the NFH-Kuza-Google rural service delivery pilot.

components:
  schemas:

    # ---- Sub-schemas ----

    QuantitySpec:
      type: object
      properties:
        value:
          type: number
        unit:
          type: string
        description:
          type: string

    ServiceArea:
      type: object
      required: [countyCode, countyName]
      properties:
```

```yaml
        countyCode:
          type: string
        countyName:
          type: string
        subCounties:
          type: array
          items:
            type: object
            required: [subCountyCode, subCountyName]
            properties:
              subCountyCode:
                type: string
              subCountyName:
                type: string

    FarmerLocation:
      type: object
      required: [county, subCounty]
      properties:
        county:
          type: string
        subCounty:
          type: string
        ward:
          type: string
        village:
          type: string
        gpsCoordinates:
          type: object
          properties:
            latitude:
              type: number
            longitude:
              type: number

    Feedback:
      type: object
      properties:
        rating:
          type: integer
          minimum: 1
          maximum: 5
        selectedOption:
          type: string
        additionalText:
          type: string

    # ---- Item Attributes ----

    LivelihoodItemAttributes:
      type: object
      required: [subCategory, serviceName, serviceMode]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        serviceMode:
          type: string
          enum: [ON_SITE_VISIT, PROVIDER_LOCATION, INPUT_DELIVERY]
        quantityRequired: { $ref: "#/components/schemas/QuantitySpec" }
        estimatedDuration: { type: string, format: duration }
        inputDetails:
          type: object
          properties:
            inputName: { type: string }
            unit: { type: string }
            quantityPerUnit: { type: number }
        sectorContext:
          type: object
          properties:
```

```yaml
            livestockType: { type: string }
            breed: { type: string }
            cropType: { type: string }
            tradeType: { type: string }
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    HouseholdItemAttributes:
      type: object
      required: [subCategory, serviceName, serviceMode]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        serviceMode:
          type: string
          enum: [SERVICE_DELIVERY, PRODUCT_DELIVERY, FACILITY_VISIT, DIGITAL]
        quantityRequired: { $ref: "#/components/schemas/QuantitySpec" }
        beneficiaryType:
          type: string
          enum: [INDIVIDUAL, HOUSEHOLD, CHILD, ELDERLY]
        documentsRequired:
          type: array
          items: { type: string }
        serviceFrequency:
          type: string
          enum: [ONE_TIME, RECURRING, SEASONAL]
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    MarketProduceItemAttributes:
      type: object
      required: [subCategory, serviceName, produceCategory, unitOfMeasure, collectionMode]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        produceCategory: { type: string }
        acceptedGrades:
          type: array
          items: { type: string }
        unitOfMeasure: { type: string }
        minimumQuantity: { type: number }
        quantityAvailable:
          type: object
          properties:
            value: { type: number }
            unit: { type: string }
        collectionMode:
          type: string
          enum: [FARM_GATE_PICKUP, COLLECTION_POINT, FARMER_DELIVERS]
        paymentTimeline:
          type: string
          enum: [IMMEDIATE, UPON_COLLECTION, AFTER_SALE]
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
```

```yaml
      tags:
        type: array
        items: { type: string }

    AdvisoryContentItemAttributes:
      type: object
      required: [subCategory, serviceName, contentType, contentUrl]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        contentType:
          type: string
          enum: [TEXT, VIDEO, PDF, AUDIO, IMAGE, INFOGRAPHIC]
        contentMimeType: { type: string }
        contentUrl: { type: string, format: uri }
        contentDuration: { type: string, format: duration }
        targetAudience: { type: string }
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    # ---- Offer Attributes ----

    LocalServiceOfferAttributes:
      type: object
      required: [priceAmount, currency, pricingModel, acceptedPaymentMethods]
      properties:
        priceAmount: { type: number }
        currency: { type: string, pattern: "^[A-Z]{3}$" }
        priceUnit: { type: string }
        pricingModel:
          type: string
          enum: [FIXED, PER_UNIT, PER_VISIT, NEGOTIABLE]
        acceptedPaymentMethods:
          type: array
          items: { type: string, enum: [CASH, MPESA] }
          minItems: 1
        capacity:
          type: object
          properties:
            value: { type: number }
            unit: { type: string }
            period: { type: string }
        serviceAvailability:
          type: object
          properties:
            availableDays:
              type: array
              items: { type: string }
            timeSlots:
              type: array
              items: { type: string }
        cancellationPolicy:
          type: object
          properties:
            isCancellable: { type: boolean }
            windowHours: { type: integer }
            reasons:
              type: array
              items: { type: string }
        validFrom: { type: string, format: date-time }
        validTo: { type: string, format: date-time }

    # ---- Fulfillment Attributes ----

    LocalServiceFulfillmentAttributes:
```

```yaml
      type: object
      required: [deliveryMode, serviceStatus]
      properties:
        serviceArea:
          type: array
          items: { $ref: "#/components/schemas/ServiceArea" }
        farmerLocation:
          $ref: "#/components/schemas/FarmerLocation"
        scheduledDate: { type: string, format: date }
        scheduledTimeSlot: { type: string }
        confirmedVisitTime: { type: string, format: date-time }
        actualServiceDate: { type: string, format: date }
        deliveryMode:
          type: string
          enum: [ON_SITE, PROVIDER_LOCATION, COLLECTION_POINT, DIGITAL]
        serviceStatus:
          type: string
          enum:
            - ORDER_PLACED
            - PROVIDER_ACCEPTED
            - PROVIDER_REJECTED
            - IN_PROGRESS
            - COMPLETED
            - FARMER_CONFIRMED
            - CANCELLED
            - CLOSED
        providerNotes: { type: string }
        followUpRequired: { type: boolean }

    # ---- Order Attributes ----

    LocalServiceOrderAttributes:
      type: object
      properties:
        farmerPaymentMethod:
          type: string
          enum: [CASH, MPESA]
        farmerPaymentStatus:
          type: string
          enum: [PAID, NOT_PAID]
        providerPaymentReceived: { type: boolean }
        providerAmountReceived: { type: number }
        paymentCurrency: { type: string, pattern: "^[A-Z]{3}$" }
        serviceConfirmedByFarmer: { type: boolean }
        providerFulfillmentStatus:
          type: string
          enum: [COMPLETED, NOT_COMPLETED]
        providerFeedback: { $ref: "#/components/schemas/Feedback" }
        farmerFeedback: { $ref: "#/components/schemas/Feedback" }
        followUpRequired: { type: boolean }
        followUpNotes: { type: string }
```

Full OpenAPI 3.1 Bundle — `health-services`

The complete `health-services` schema definitions bundled as a single OpenAPI 3.1 components block. Shared sub-schemas (`QuantitySpec`, `ServiceArea`, `SeekerLocation`, `Feedback`) are reused from the `local-services` bundle above.

```yaml
# ============================================================
# NFH-Kuza-Google Extension Schemas
# Domain: health-services (TIBU) | Protocol: Beckn v2.0.0
# ============================================================

openapi: "3.1.0"
info:
  title: NFH-Kuza-Google Health Extension Schemas
  version: "1.0.0"
  description: >
    Domain extension schemas for the health-services domain
    (TIBU / Tuberculosis) in the NFH-Kuza-Google platform.
    Shared sub-schemas are imported from the local-services bundle.
```

```yaml
components:
  schemas:

    # ---- Health Sub-schemas ----

    AdverseEventDetails:
      type: object
      required: [eventType, severity]
      properties:
        eventType: { type: string }
        severity:
          type: string
          enum: [MILD, MODERATE, SEVERE, LIFE_THREATENING]
        onsetDate: { type: string, format: date }
        action: { type: string }
        outcome: { type: string }

    ReferralTarget:
      type: object
      required: [facilityId, facilityName]
      properties:
        facilityId: { type: string }
        facilityName: { type: string }
        reason: { type: string }
        urgency:
          type: string
          enum: [ROUTINE, URGENT, EMERGENCY]

    SeekerLocation:
      type: object
      required: [county, subCounty]
      properties:
        county: { type: string }
        subCounty: { type: string }
        ward: { type: string }
        village: { type: string }
        gpsCoordinates:
          type: object
          properties:
            latitude: { type: number }
            longitude: { type: number }

    # ---- Health Item Attributes ----

    HealthScreeningItemAttributes:
      type: object
      required: [subCategory, screeningType, diseaseTarget]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        screeningType:
          type: string
          enum: [SYMPTOM_SCREENING, SPUTUM_COLLECTION, XRAY_SCREENING, CONTACT_TRACING, OTHER]
        diseaseTarget: { type: string }
        diagnosticMethod: { type: string }
        requiresLabCapability:
          type: array
          items: { type: string }
        patientEligibility:
          type: object
          properties:
            minAge: { type: integer }
            maxAge: { type: integer }
            riskFactors:
              type: array
              items: { type: string }
        expectedTurnaroundHours: { type: integer }
        isFreeAtPointOfCare: { type: boolean }
        availableLanguages:
          type: array
```

```yaml
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    HealthTreatmentItemAttributes:
      type: object
      required: [subCategory, treatmentType]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        treatmentType:
          type: string
          enum: [DSTB, DRTB, TPT, OTHER]
        regimenName: { type: string }
        treatmentDurationDays: { type: integer }
        frequency: { type: string }
        requiresMonitoring:
          type: array
          items: { type: string }
        adherenceTracking: { type: boolean }
        adverseEventReporting: { type: boolean }
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    HealthSupervisionItemAttributes:
      type: object
      required: [subCategory, supervisionType, targetEntity]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        supervisionType:
          type: string
          enum: [SCTLC_VISIT, CTLC_VISIT, DATA_QUALITY_AUDIT, MORTALITY_AUDIT, MENTORSHIP]
        targetEntity:
          type: string
          enum: [FACILITY, CHP, CLINICIAN, LAB]
        frequency: { type: string }
        checklistTemplate: { type: string, format: uri }
        includesPaymentClaim: { type: boolean }
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    HealthAdvisoryItemAttributes:
      type: object
      required: [subCategory, contentType, contentUrl]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        contentType:
          type: string
          enum: [TEXT, VIDEO, PDF, AUDIO, INFOGRAPHIC, INTERACTIVE]
        contentMimeType: { type: string }
        contentUrl: { type: string, format: uri }
        contentDuration: { type: string, format: duration }
```

```yaml
      targetAudience: { type: string }
      healthTopic: { type: string }
      availableLanguages:
        type: array
        items: { type: string, pattern: "^[a-z]{2}$" }
      tags:
        type: array
        items: { type: string }

    # ---- Health Offer Attributes ----

    HealthServiceOfferAttributes:
      type: object
      properties:
        priceAmount: { type: number }
        currency: { type: string, pattern: "^[A-Z]{3}$" }
        isFreeAtPointOfCare: { type: boolean }
        coveredBy:
          type: string
          enum: [GOVERNMENT_PROGRAM, INSURANCE, OUT_OF_POCKET, NGO_FUNDED, GLOBAL_FUND]
        coverageNote: { type: string }
        acceptedPaymentMethods:
          type: array
          items: { type: string, enum: [CASH, MPESA, INSURANCE_CLAIM, NONE] }
        capacity:
          type: object
          properties:
            value: { type: number }
            unit: { type: string }
            period: { type: string }
        serviceAvailability:
          type: object
          properties:
            availableDays:
              type: array
              items: { type: string }
            timeSlots:
              type: array
              items: { type: string }
        validFrom: { type: string, format: date-time }
        validTo: { type: string, format: date-time }

    # ---- Health Fulfillment Attributes ----

    HealthServiceFulfillmentAttributes:
      type: object
      required: [deliveryMode, serviceStatus]
      properties:
        facilityServiceArea:
          type: array
          items: { $ref: "#/components/schemas/ServiceArea" }
        patientLocation: { $ref: "#/components/schemas/SeekerLocation" }
        facilityType:
          type: string
          enum: [DISPENSARY, HEALTH_CENTRE, HOSPITAL, LAB, COMMUNITY, MOBILE]
        requiredCapabilities:
          type: array
          items: { type: string }
        providerCredentialRequired: { type: string }
        scheduledDate: { type: string, format: date }
        scheduledTimeSlot: { type: string }
        confirmedVisitTime: { type: string, format: date-time }
        actualServiceDate: { type: string, format: date }
        deliveryMode:
          type: string
          enum: [FACILITY_VISIT, HOME_VISIT, TELE_HEALTH, MOBILE_CLINIC]
        serviceStatus:
          type: string
          enum:
            - NEED_REGISTERED
            - SEARCHED
            - SELECTED
            - CONFIRMED
            - DELIVERED
            - RECORDED
```

```
            - OUTCOME_PENDING
            - OUTCOME_RECORDED
            - CANCELLED
            - LOST_TO_FOLLOWUP
            - CLOSED
        providerNotes: { type: string }
        followUpRequired: { type: boolean }
        referralRequired: { type: boolean }
        referralTarget: { $ref: "#/components/schemas/ReferralTarget" }

    # ---- Health Order Attributes ----

    HealthServiceOrderAttributes:
      type: object
      properties:
        patientUuid: { type: string }
        providerCredentialId: { type: string }
        clinicalRecordRef: { type: string }
        serviceConfirmedByPatient: { type: boolean }
        providerFulfillmentStatus:
          type: string
          enum: [COMPLETED, PARTIAL, NOT_COMPLETED]
        outcomeRecorded: { type: boolean }
        outcomeType:
          type: string
          enum: [TB_POSITIVE, TB_NEGATIVE, CURED, TREATMENT_COMPLETED, DIED, LOST_TO_FOLLOWUP, TRANSFERRED_OUT, TREATMENT_FAILED,
NOT_EVALUATED]
        outcomeDate: { type: string, format: date }
        outcomeNote: { type: string }
        adverseEventReported: { type: boolean }
        adverseEventDetails: { $ref: "#/components/schemas/AdverseEventDetails" }
        followUpRequired: { type: boolean }
        followUpScheduledDate: { type: string, format: date }
        consentGiven: { type: boolean }
        consentTimestamp: { type: string, format: date-time }
```